A

STM Exercise 8

Phase Diagrams

(Fe-C)

IRON-CARBON (Fe-C) PHASE DIAGRAM

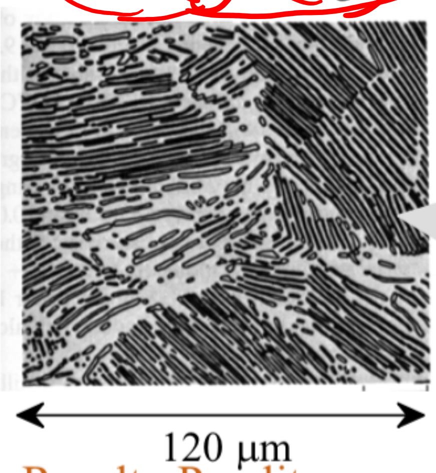
- 2 important points

  -Eutectic (A):

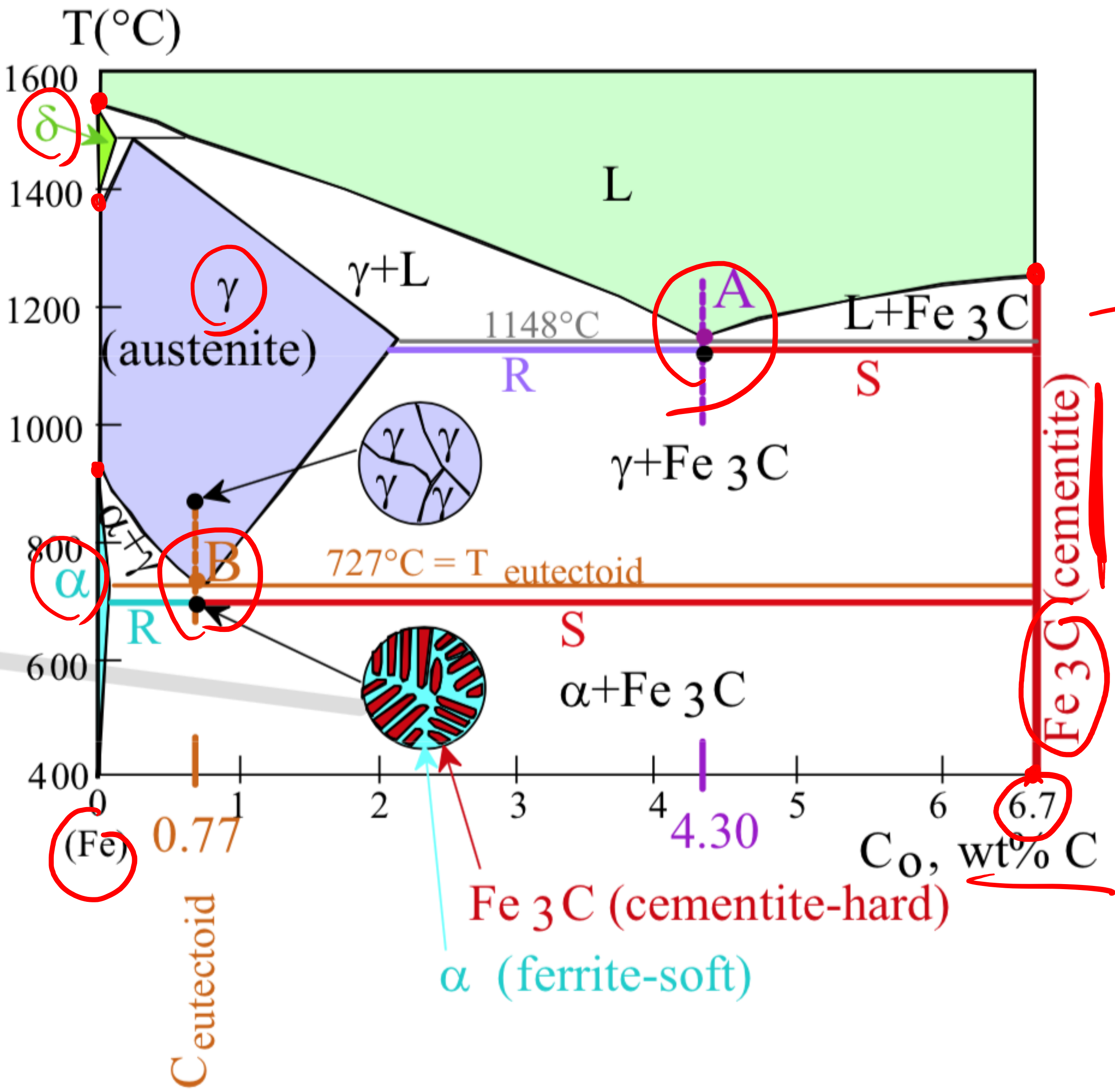
  $L \Rightarrow \gamma + Fe_3C$

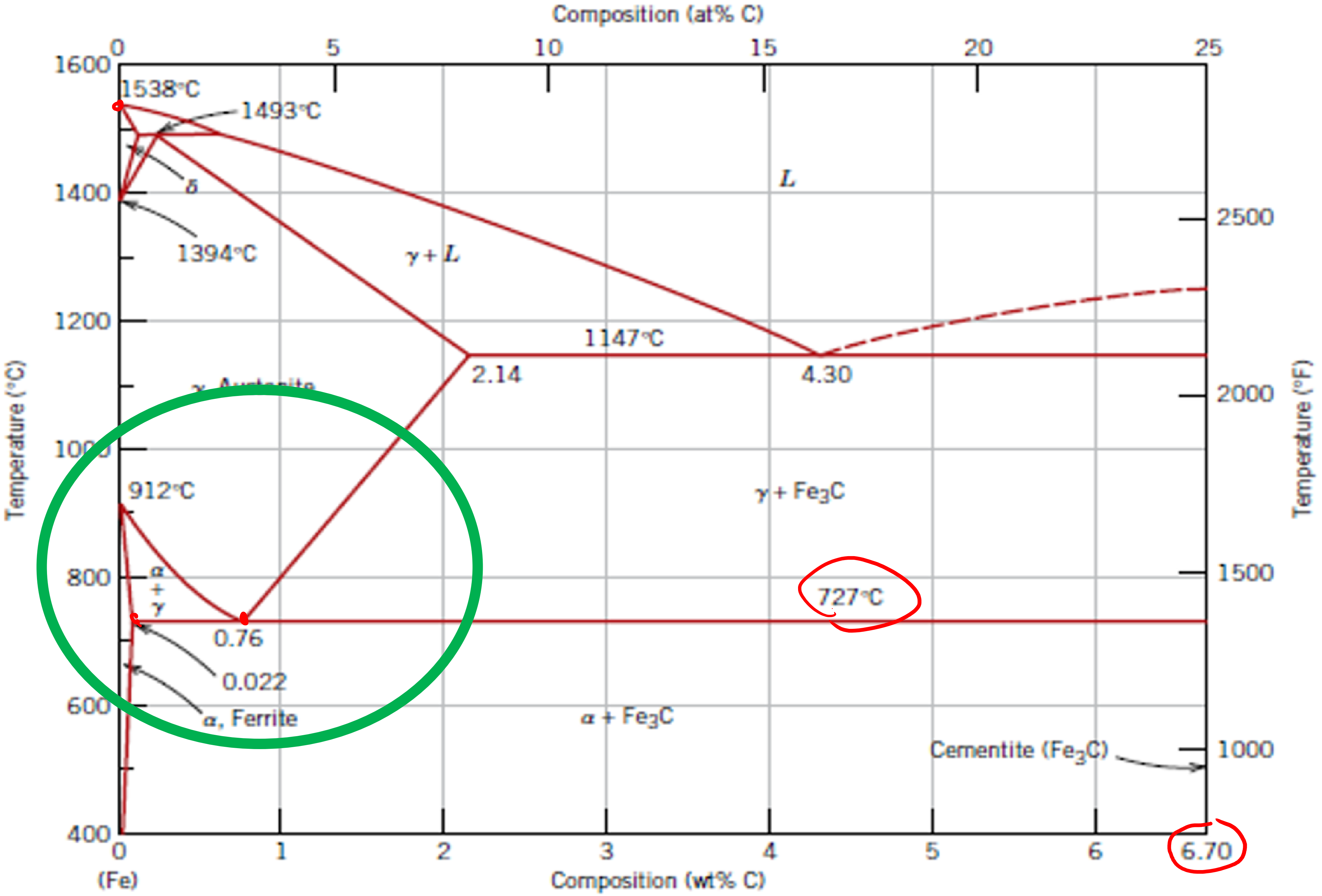
  -Eutectoid (B):

  $\gamma \Rightarrow \alpha + Fe_3C$

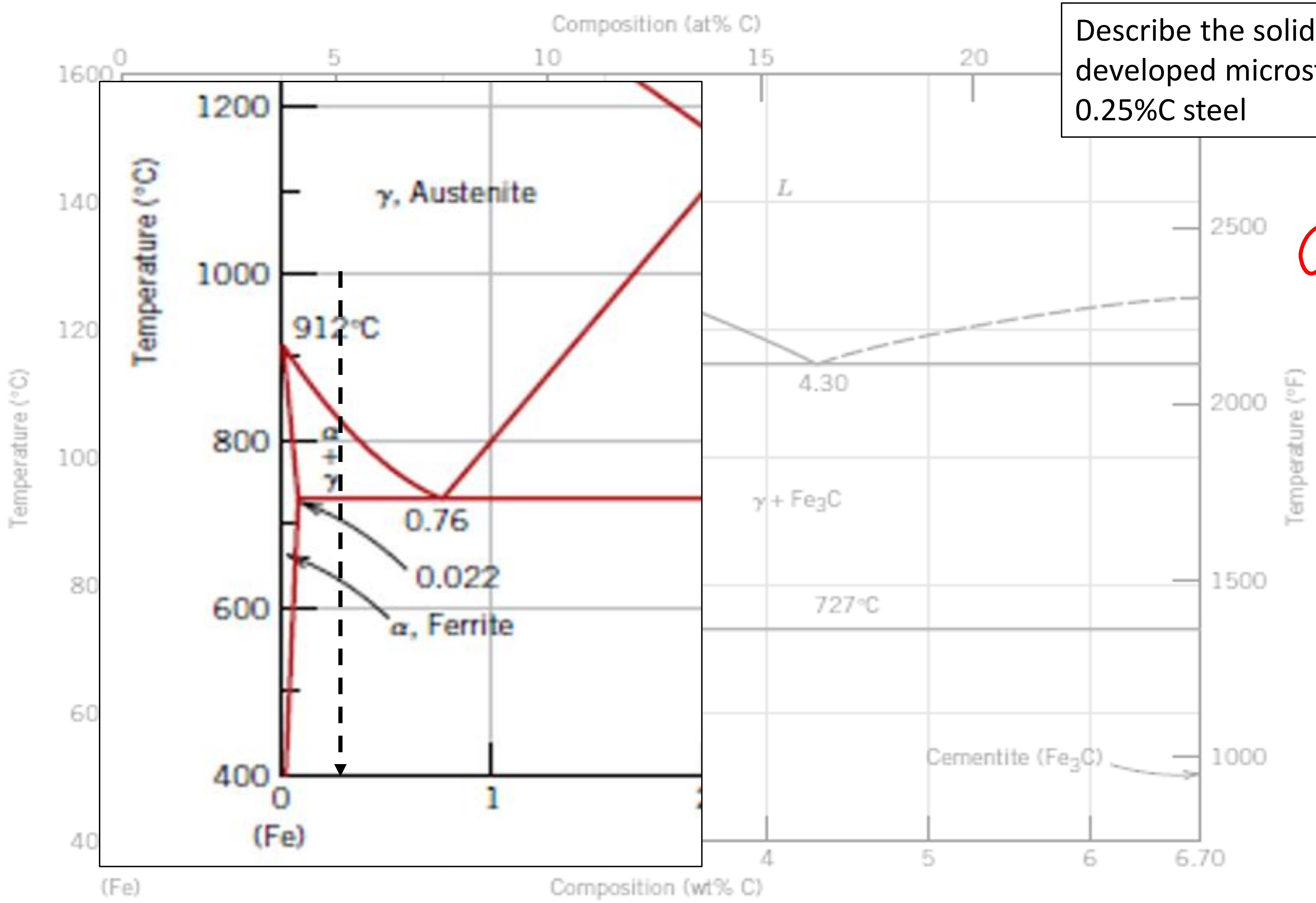


Result: Pearlite = alternating layers of α and Fe_3C phases.



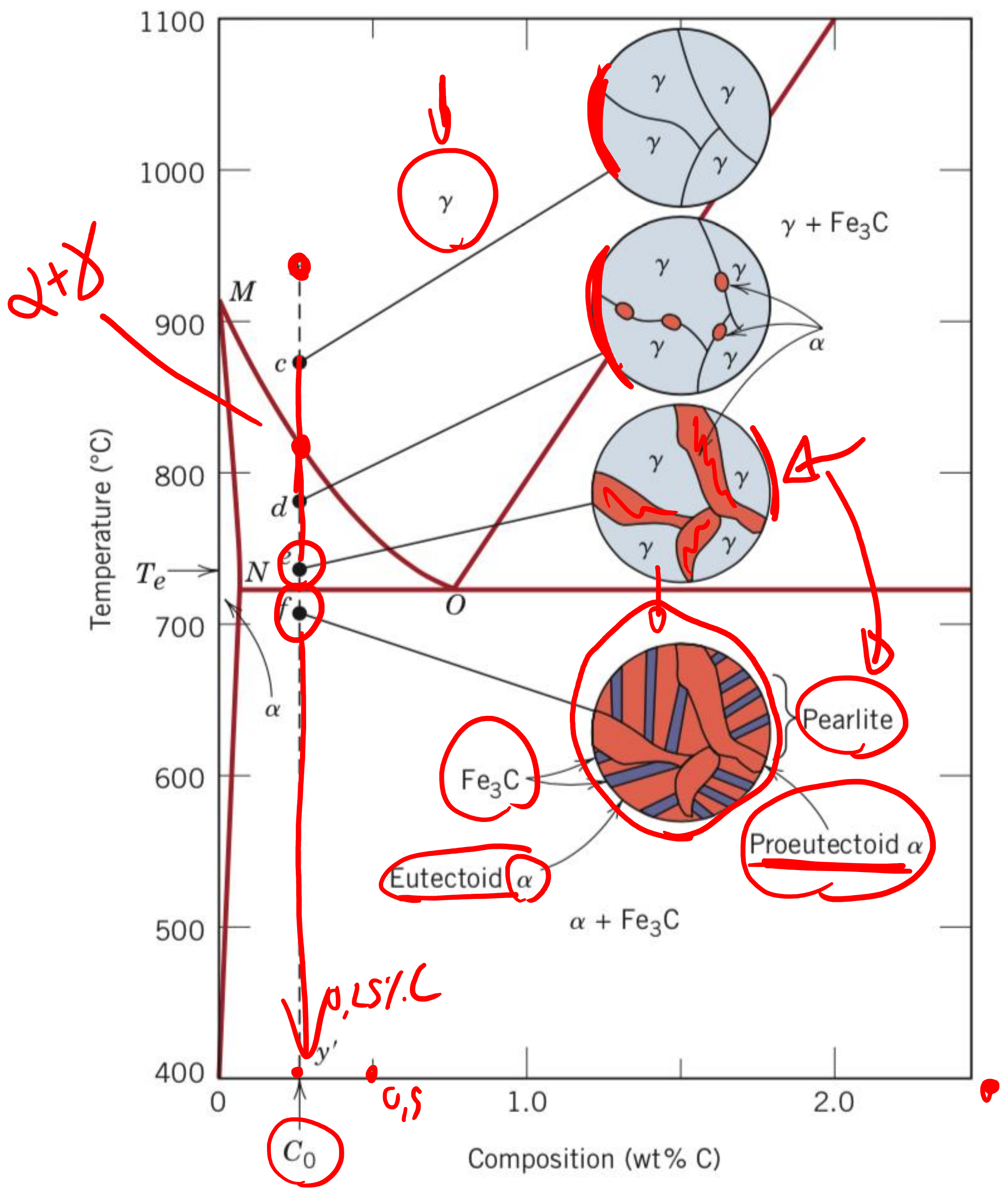


Composition (at% C)
0
5
10
15
20
25
1600
1538°C
1493°C
δ
1400
1394°C
L
γ + L
2500
1200
1147°C
2.14
4.30
γ, Austenite
2000
1000
912°C
γ + Fe3C
α
+
γ
800
727°C
1500
0.76
0.022
600
α, Ferrite
α + Fe3C
Cementite (Fe3C)
1000
400
0
(Fe)
1
2
3
4
5
6
6.70
Composition (wt% C)
Temperature (°C)
Temperature (°F)



Describe the solidification (evidencing developed microstructures) of a 0.25%C steel

$C_0 \begin{cases} 0{,}25\%\ C \\ \cancel{99{,}75\%\ Fe} \end{cases}$



α Proeutectoid alpha(ferrite)

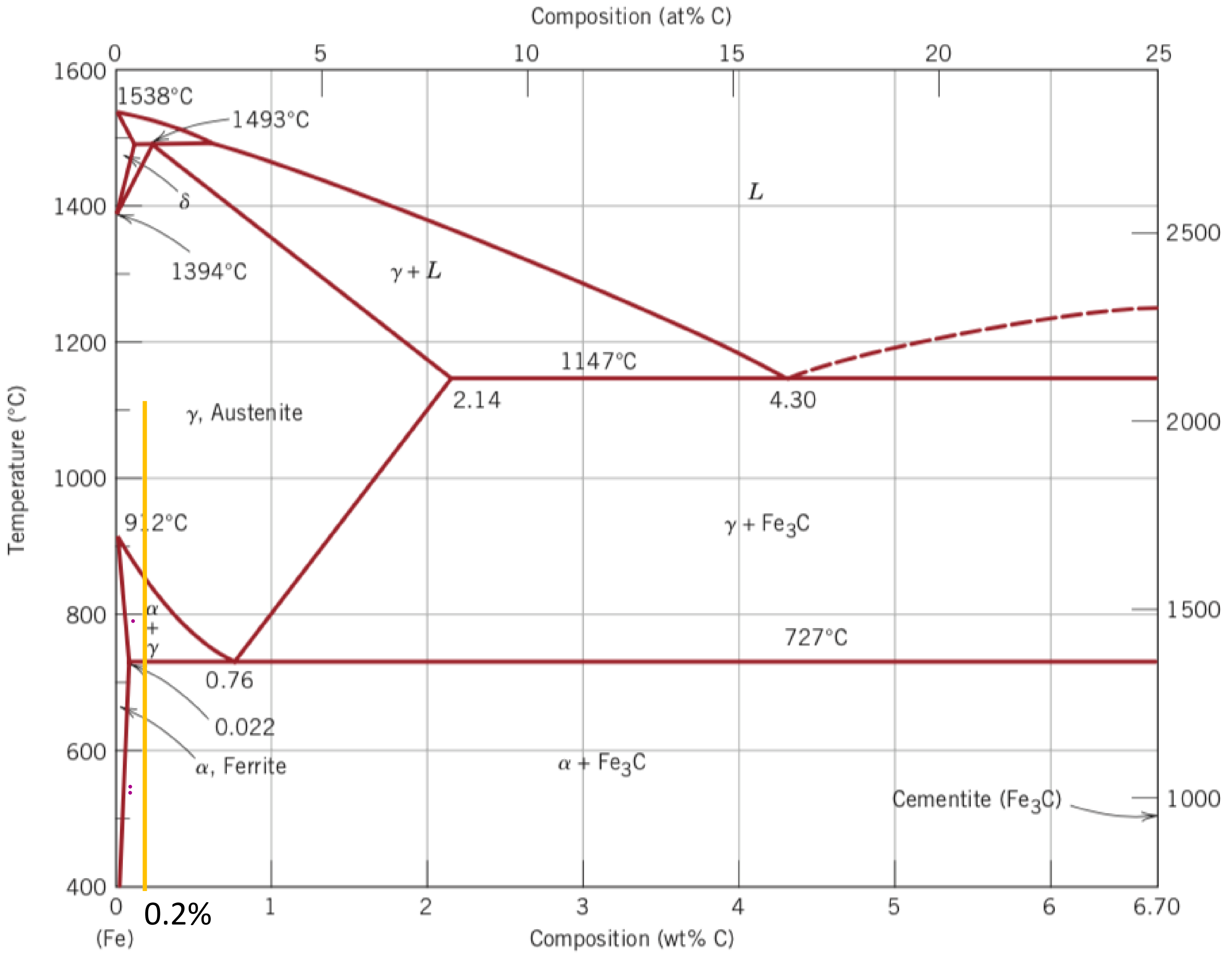
$\alpha + Fe_3C$

Eutectoid reaction: Austenite -> $\underbrace{\text{Ferrite + Cementite}}_{\text{Pearlite}}$

$\gamma \rightarrow$ Pearlite

Use the Fe-Fe_3C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) ferrite for a 0,2 % C steel



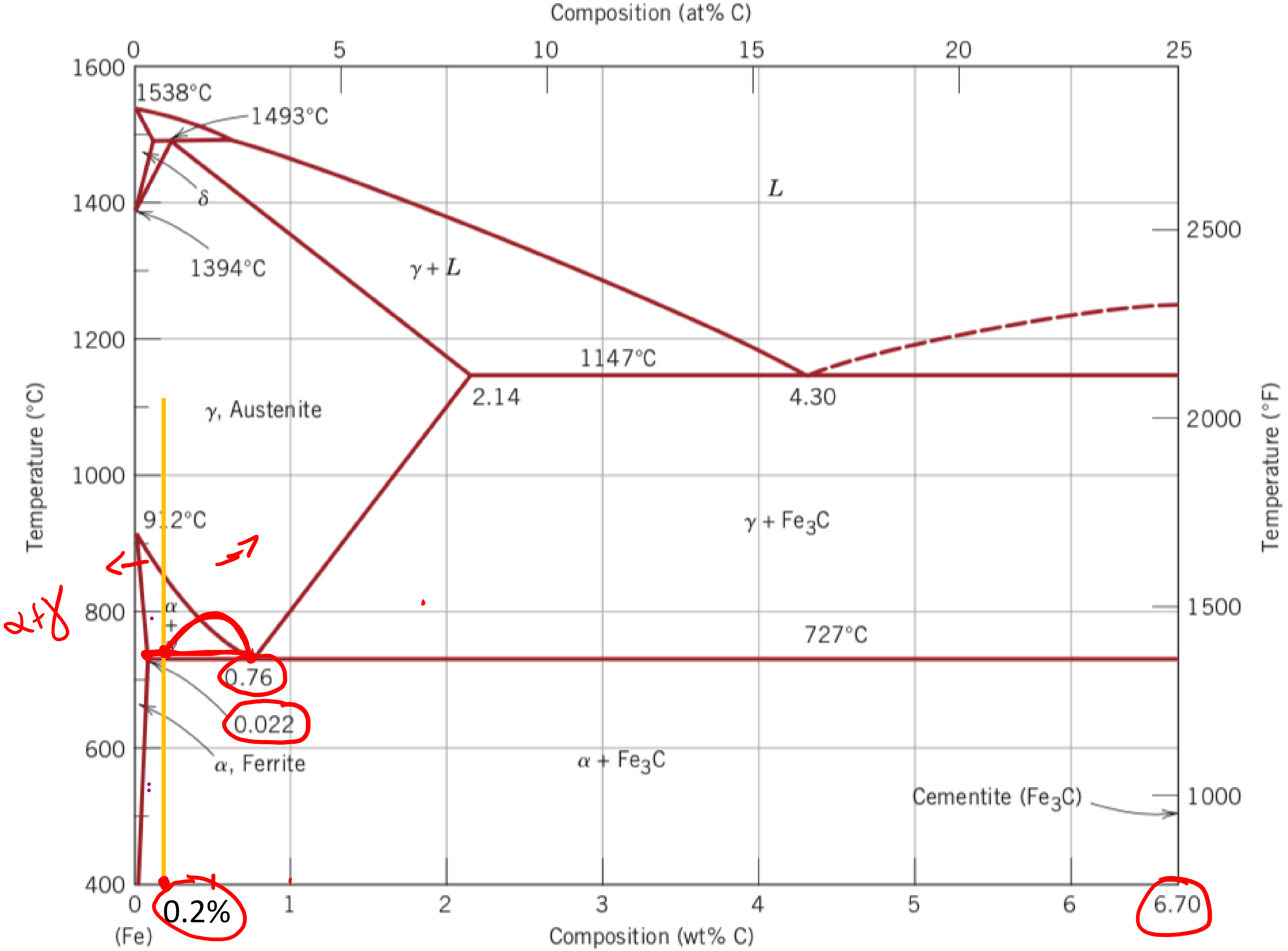
Use the Fe-Fe_3C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) ferrite for a 0,2 % C steel

$C_0 \left\{ 0{,}2\,\%\,C \right.$

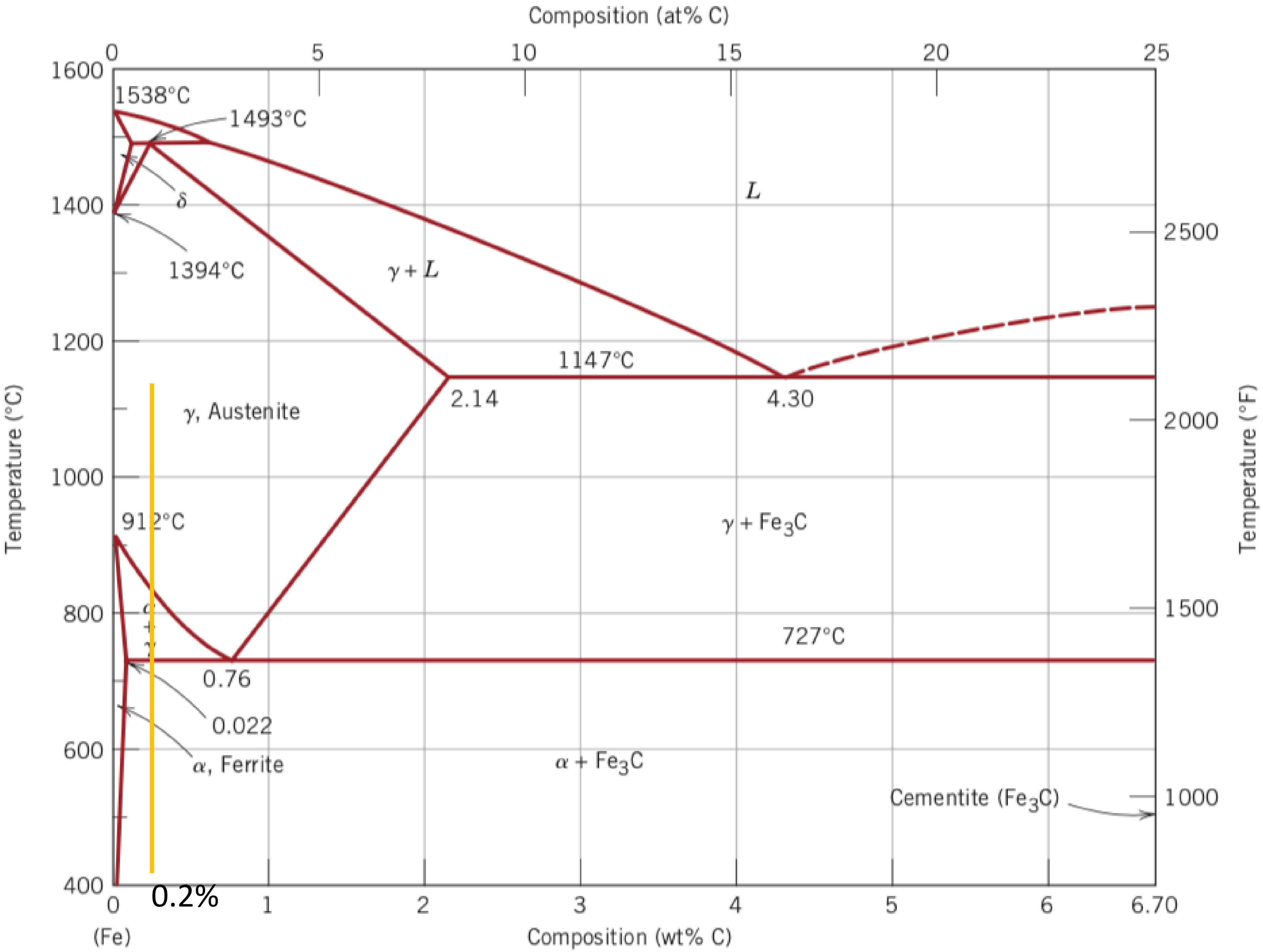
$$\alpha_p = \frac{0{,}76 - 0{,}2}{0{,}76 - 0{,}022} \cdot 100 =$$

$= 76\%$



Use the Fe-Fe3C phase diagram to calculate:

-the percentage of pearlite for a 0,2 % C steel



Use the Fe-Fe3C phase diagram to calculate:

-the percentage of pearlite for a 0,2 % C steel

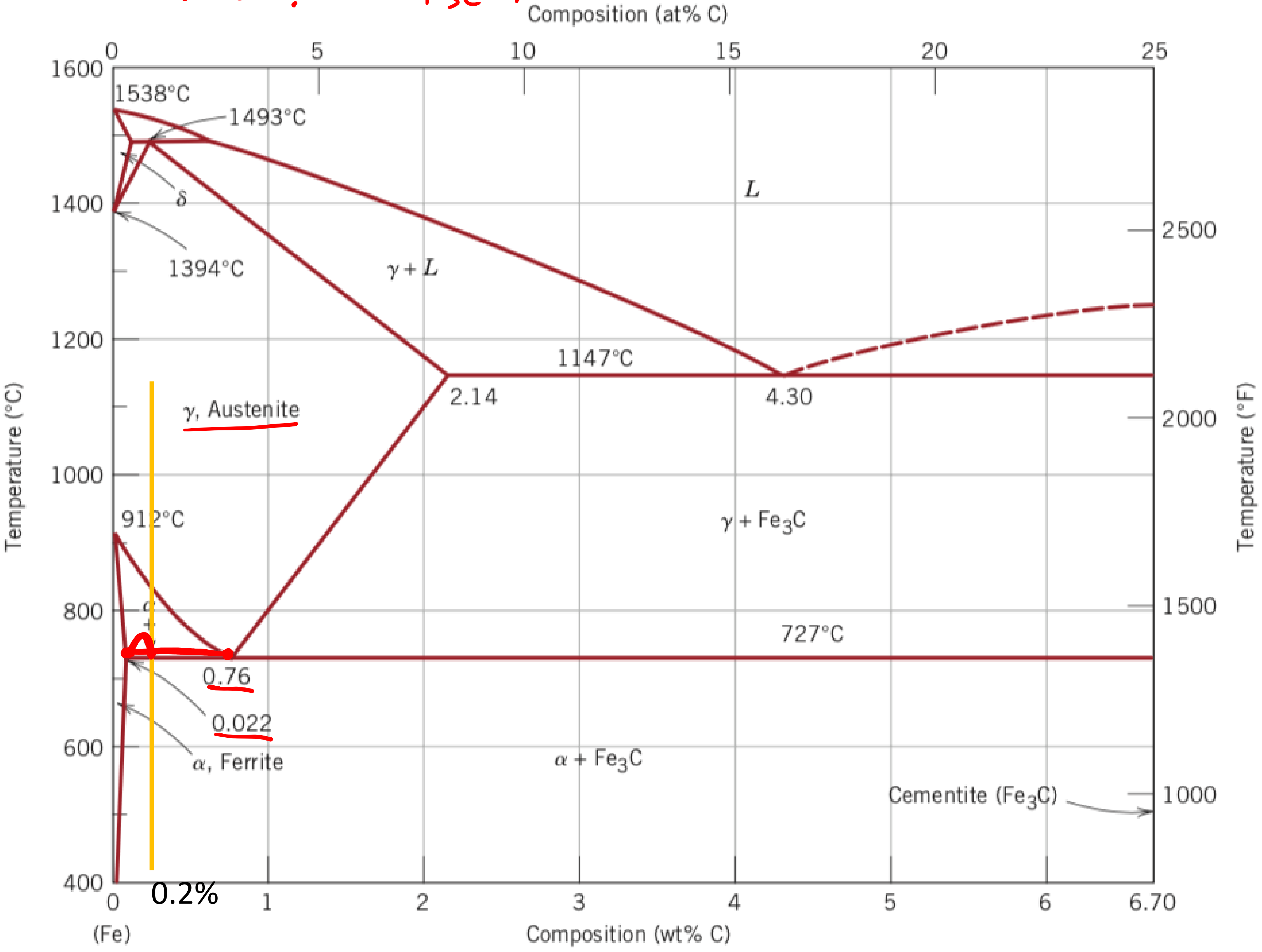
$E_{nt}\ \alpha = ?$ $\quad E_{nt}\ Fe_3C = ?$

$$Pearlite = \frac{0{,}2 - 0{,}022}{0{,}76 - 0{,}022} \cdot 100 =$$

$$= \boxed{24\%}$$

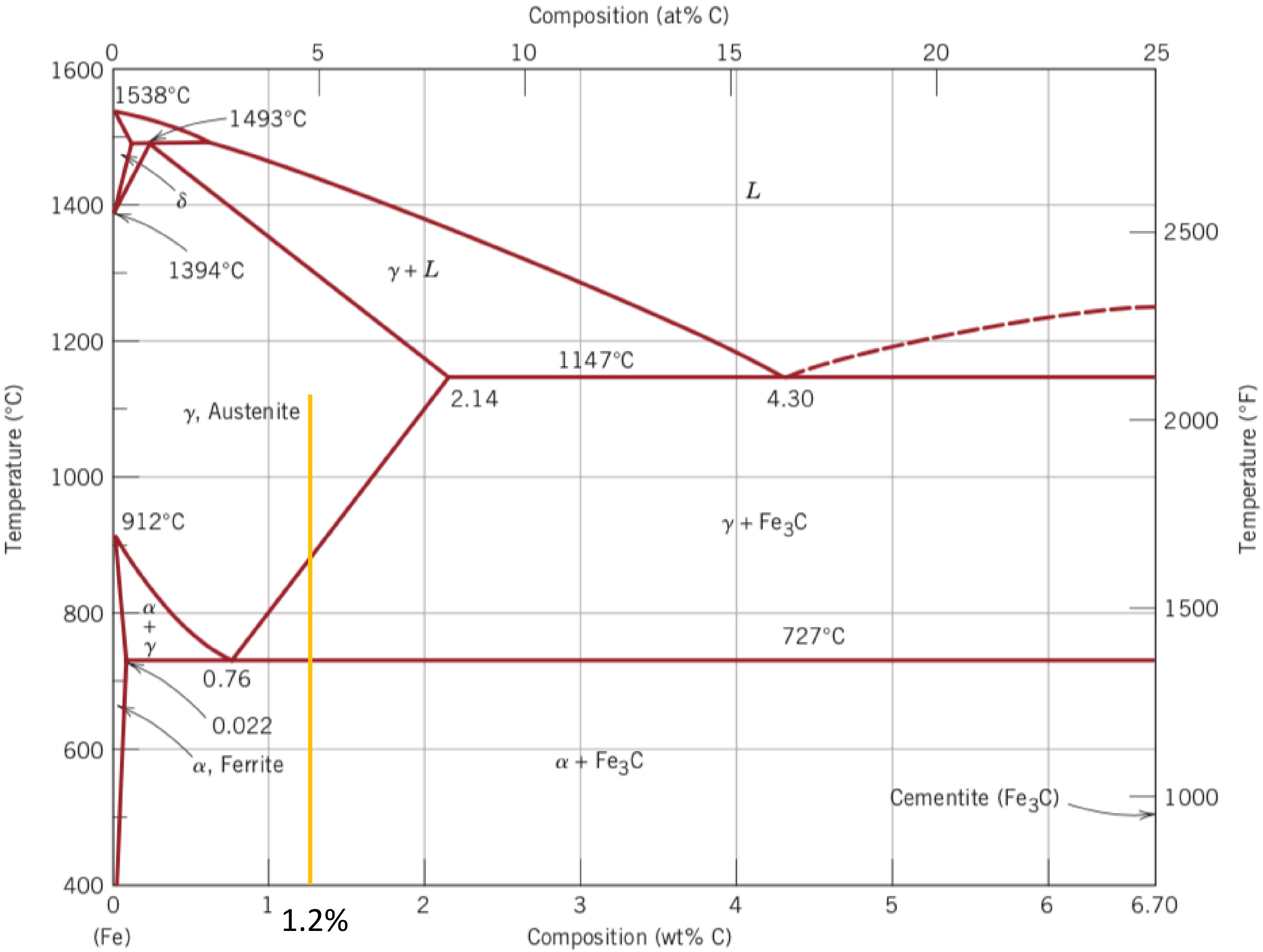
$$Pearlite = 100\% - \alpha_p =$$

$$= 100 - 76 = 24\%$$



Use the Fe-Fe3C phase diagram to calculate:

-the percentage of pearlite for a 1,2 % C steel

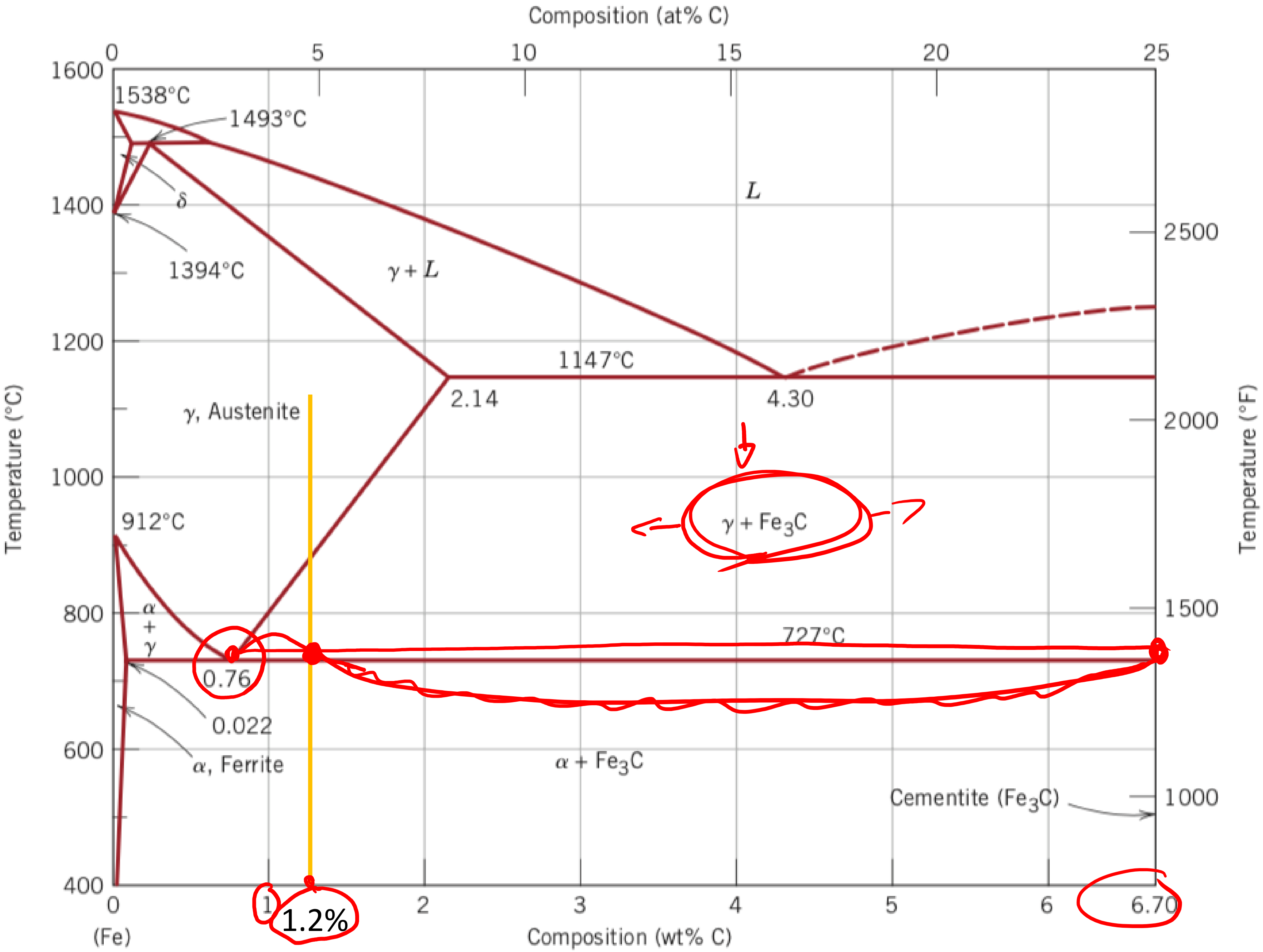


Use the Fe-Fe3C phase diagram to calculate:

-the percentage of pearlite for a 1,2 % C steel

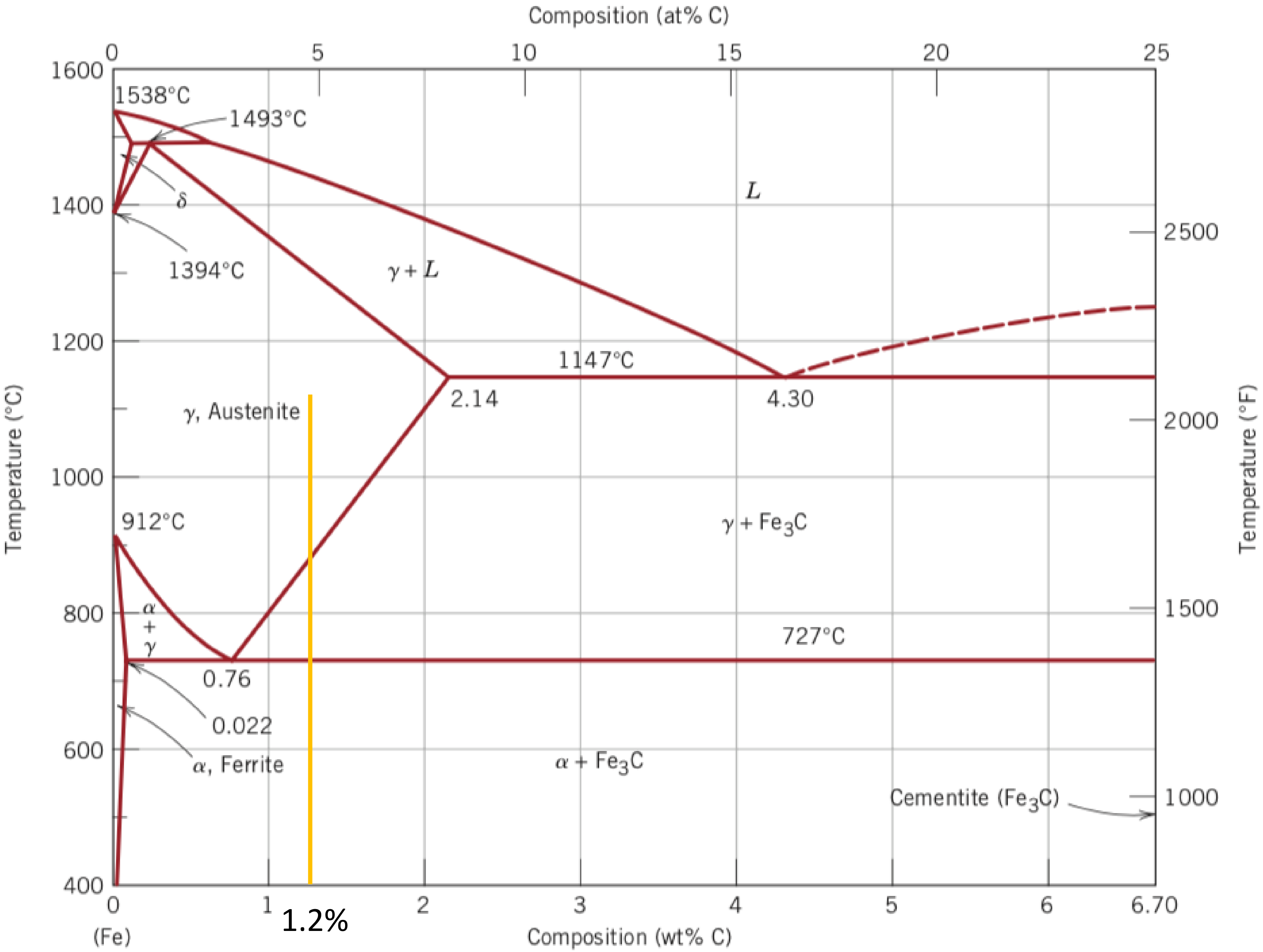
$$\text{Pearlite} = \frac{6,7 - 1,2}{6,7 - 0,76} \cdot 100 =$$

$= 92,6\ \%$



Use the Fe-Fe3C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) cementite for a 1,2 % C steel



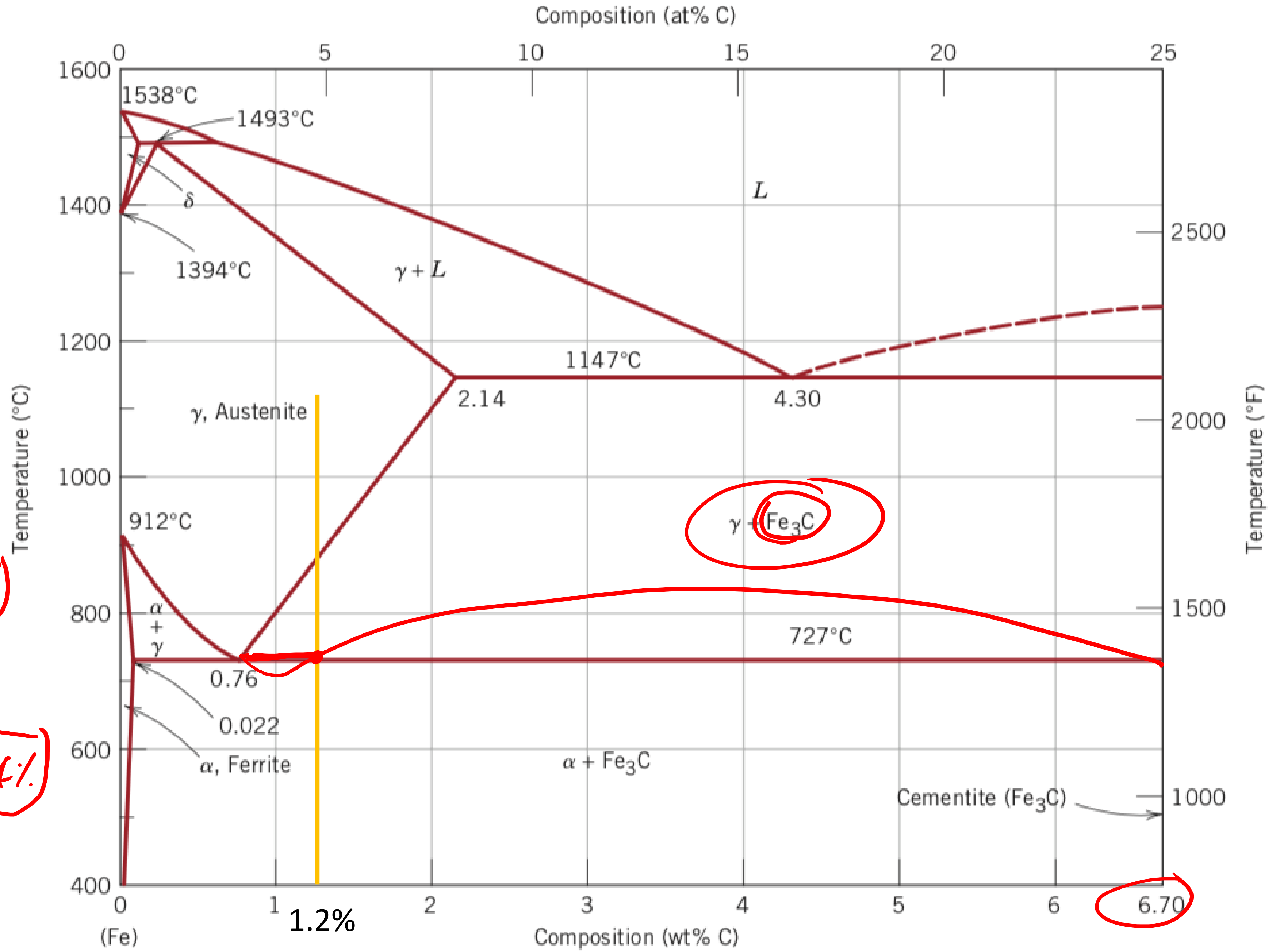
Use the Fe-Fe3C phase diagram to calculate:

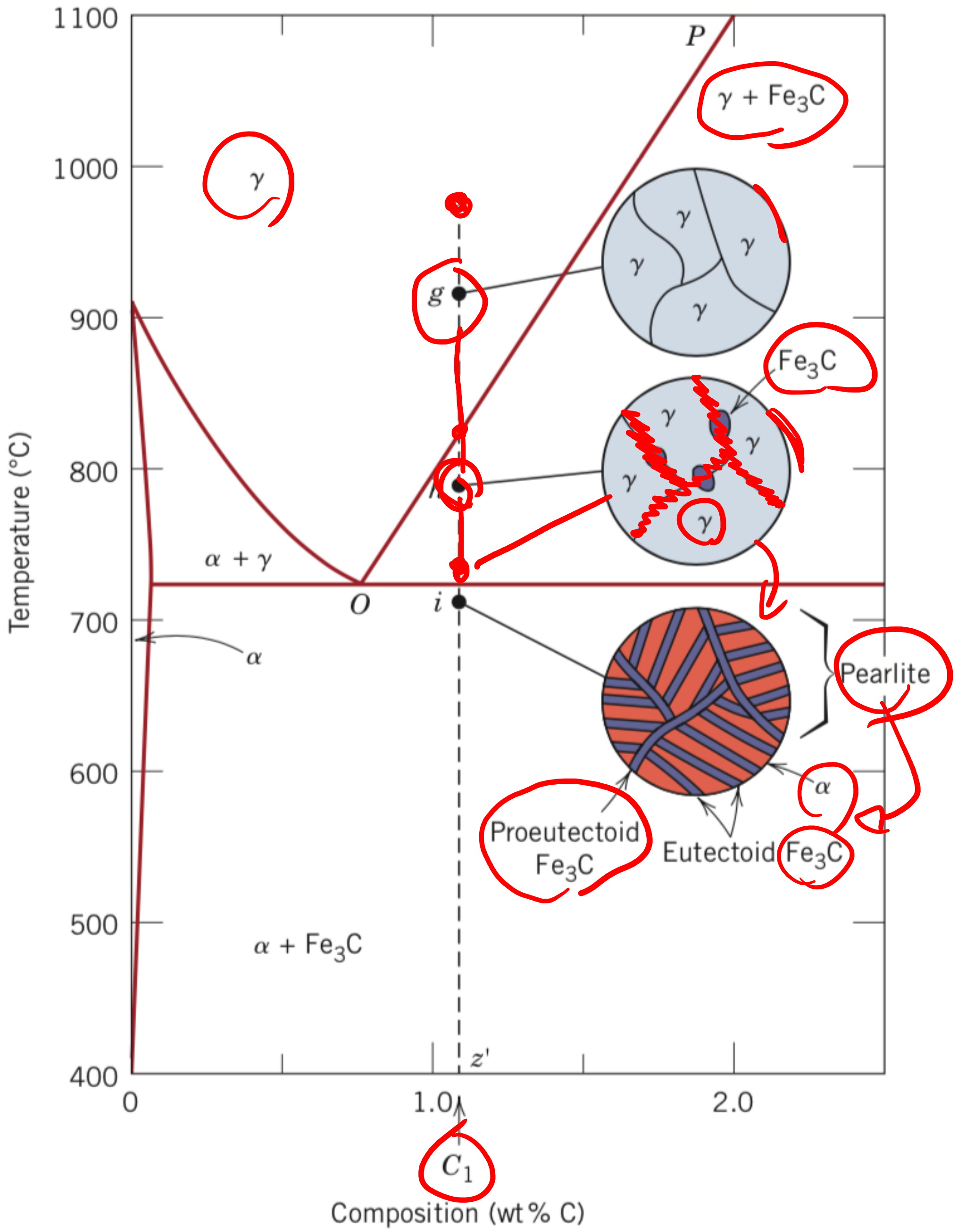
-the percentage of primary (pro-eutectoid) cementite for a 1,2 % C steel

$(Fe_3C)_p = 100\% - \text{Perlite} =$

$= 100 - 92{,}6 = 7{,}4\%$

$$(Fe_3C)_p = \frac{1{,}2 - 0{,}76}{6{,}7 - 0{,}76} \cdot 100 = 7{,}4\%$$



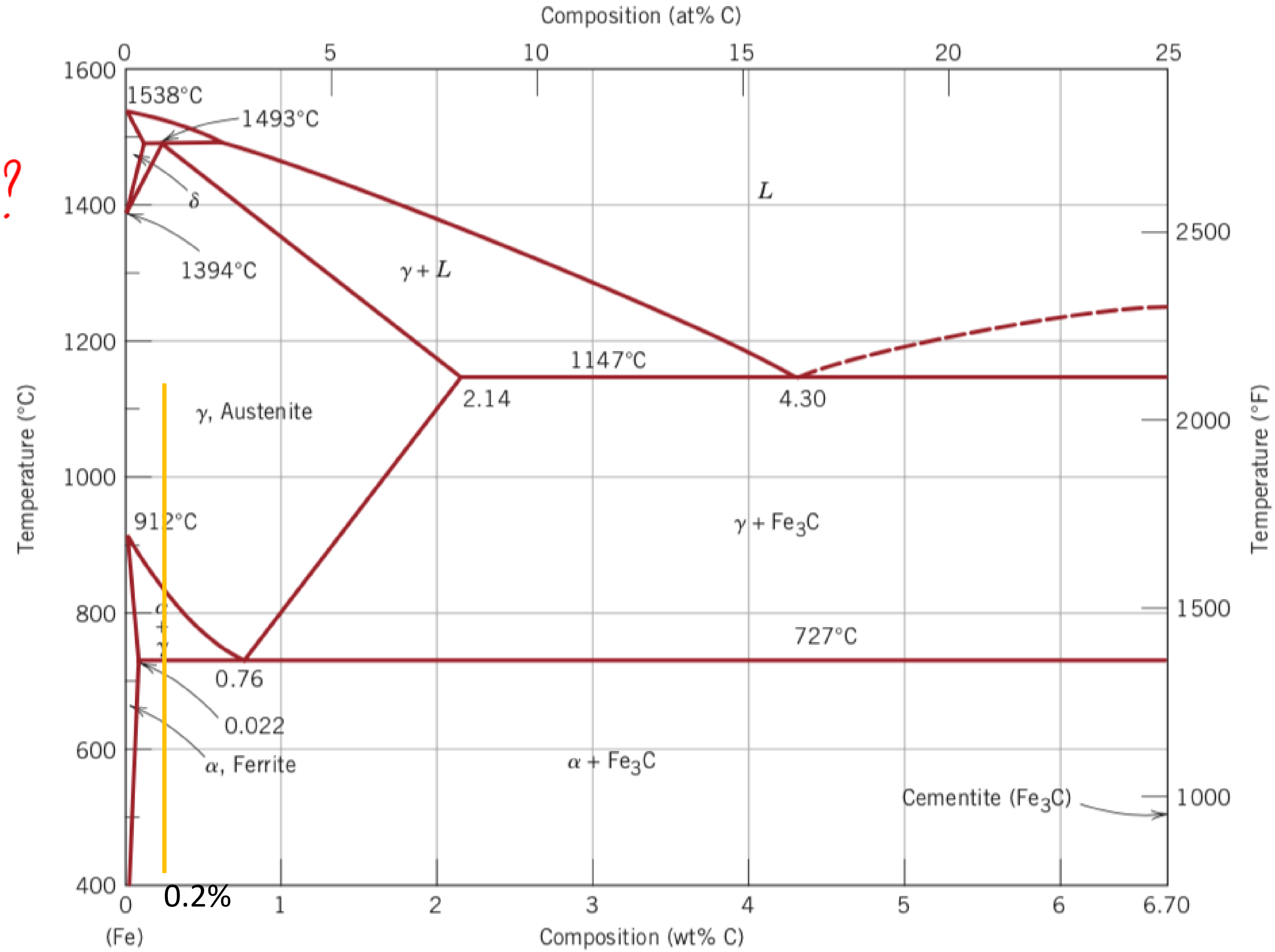


1100
1000
900
800
700
600
500
400
Temperature (°C)
0
1.0
2.0
Composition (wt% C)
C1
P
γ + Fe3C
γ
g
Fe3C
α + γ
O
i
α
Pearlite
α
Proeutectoid Fe3C
Eutectoid Fe3C
α + Fe3C
z'

Use the Fe-Fe3C phase diagram to calculate:

-the ~~percentage of pearlite for~~ a 0,2 % C steel

$Eut\ \alpha = ?$ $Eut(Fe_3C) = ?$



Use the Fe-Fe3C phase diagram to calculate:

-~~the percentage of pearlite~~ for a 0,2 % C steel

$Eut\ \alpha = ?$ $\quad Eut(Fe_3C) = ?$

$\boxed{\alpha_p} = 76\ \%$ $\quad \boxed{PEARLITE} = 24\ \%$

$$\boxed{Eut\ (Fe_3C)} = \frac{0{,}2 - 0{,}022}{6{,}7 - 0{,}022} \cdot 100$$

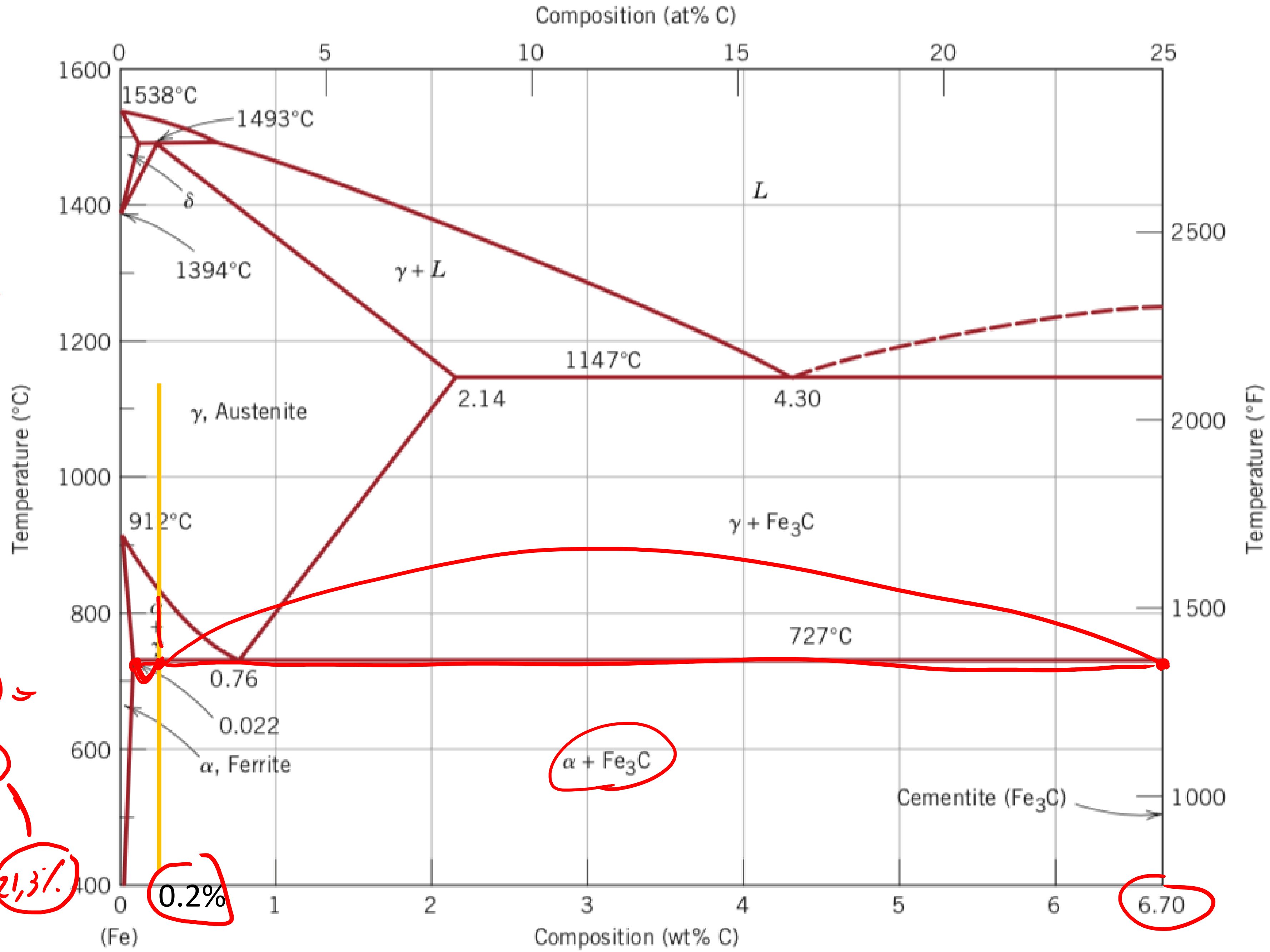
$= 2{,}7\ \%$

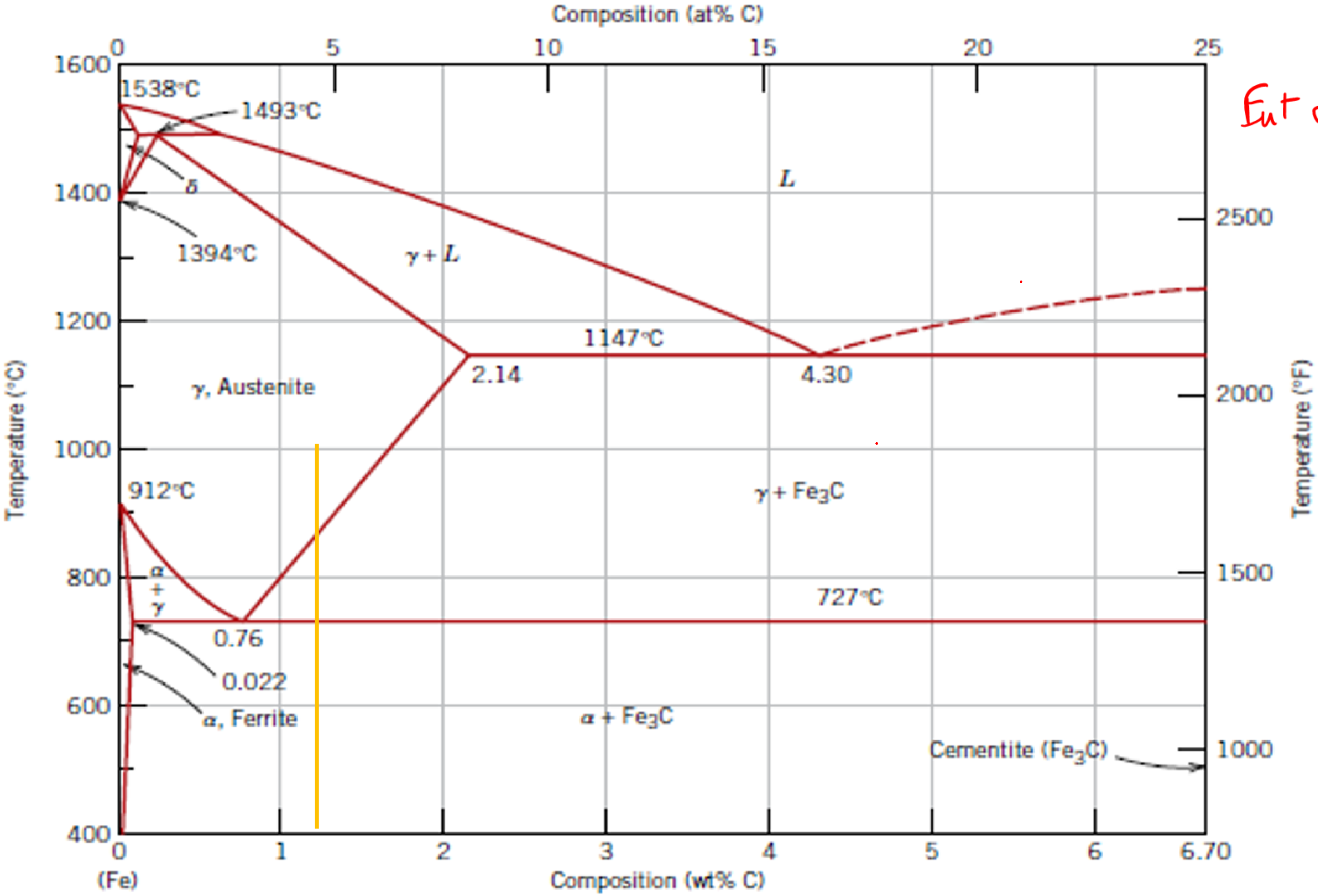
$$\boxed{Eut\ \alpha} = PEARLITE - Eut\ (Fe_3C) =$$

$= 24 - 2{,}7 = 21{,}3\ \%$

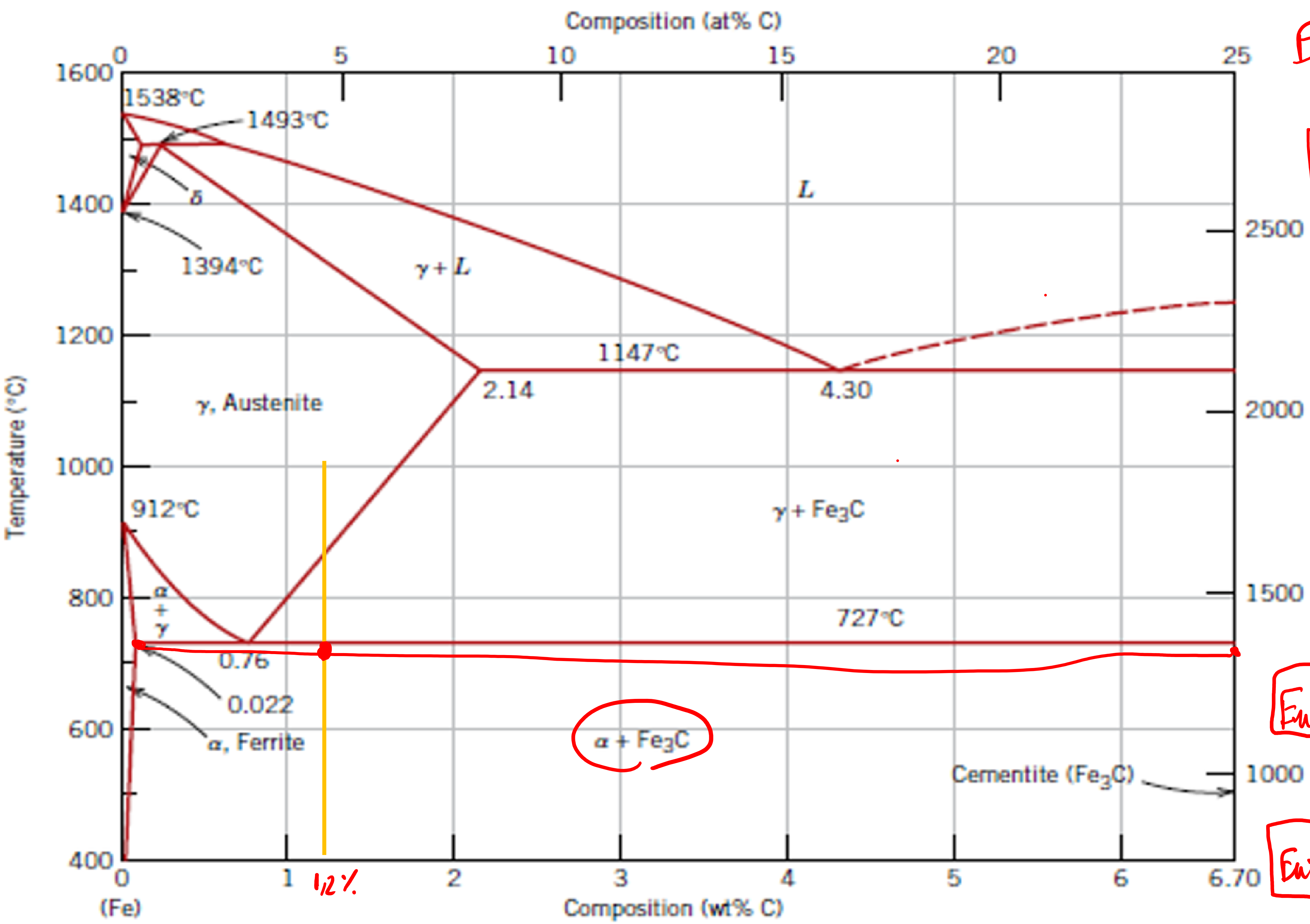
$TOT\ \alpha = 100 - 2{,}7 = 97{,}3\ \%$

$\boxed{Eut\ \alpha} = TOT\ \alpha - \alpha_p = 97{,}3 - 76 = 21{,}3\ \%$





Eut α = ? Eut $[Fe_3C]$ = ?



Ent α = ? Ent $[Fe_3C]$ = ?

$[Fe_3C]_P$ = 7,4%

PEARLITE = 92,6%

$$Ent_\alpha = \frac{6{,}7 - 1{,}2}{6{,}7 - 0{,}022} \cdot 100$$

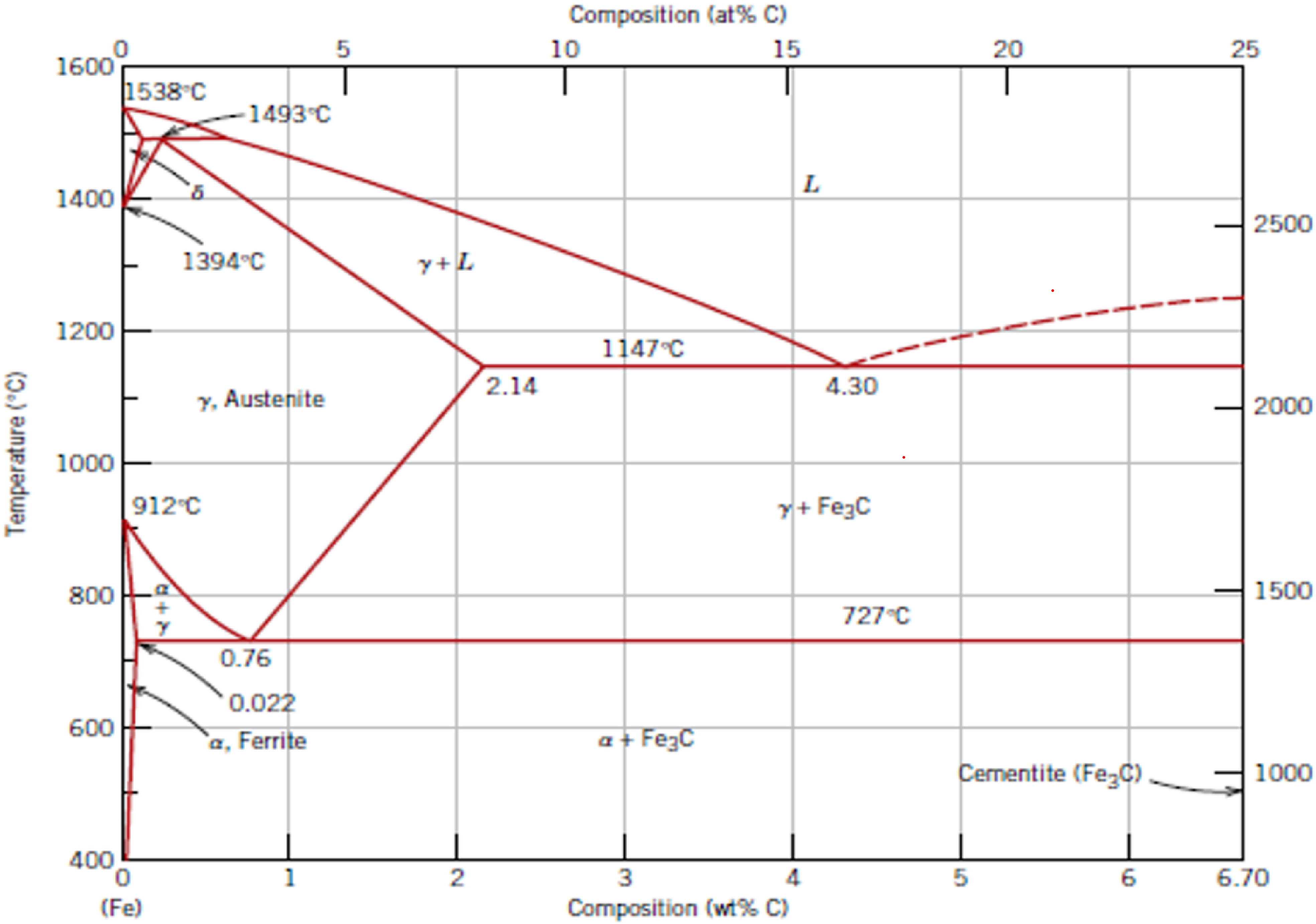
= 82,3%

TOT $[Fe_3C]$ = 17,7%

$\boxed{Ent[Fe_3C]}$ = TOT $[Fe_3C] - [Fe_3C]_P$ =

= 17,7% − 7,4% = 10,3%

$\boxed{Ent[Fe_3C]}$ = PEARL − Ent α = 10,3%



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α, Ferrite
α + Fe3C
Cementite (Fe3C)
2500
2000
1500
1000
Temperature (°C)
Temperature (°F)
0
(Fe)
1
2
3
4
5
6
6.70
Composition (wt% C)